Fiscal Year:	FY 2014	**Task Last Updated:**	FY 07/18/2017
PI Name:	Clark, Torin K. Ph.D.		
Project Title:	Predicting Sensorimotor Adaptation to Altered Gravity by Measuring Vestibular Perceptual Thresholds		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	torin.clark@colorado.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	303-915-2152
Organization Name:	University of Colorado, Boulder		
PI Address 1:	Smead Aerospace Engineering Sciences		
PI Address 2:	3775 Discovery Dr, Rm. AERO N301		
PI Web Page:			
City:	Boulder	**State:**	CO
Zip Code:	80303-7813	**Congressional District:**	2
Comments:	NOTE: PI moved to University of Colorado after NSBRI Postdoctoral Fellowship concluded in late 2015 (Ed., 9/1/17)		
Project Type:	Ground	**Solicitation / Funding Source:**	2013 NSBRI-RFA-13-01 Postdoctoral Fellowships
Start Date:	09/30/2013	**End Date:**	09/30/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Merfeld, Daniel (MENTOR/ Massachusetts Eye and Ear Infirmary)		
Grant/Contract No.:	NCC 9-58-PF03502		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

POSTDOCTORAL FELLOWSHIP

Astronauts must maintain appropriate sensorimotor function for tasks such as piloted landing or docking, as well as vehicle egress and extra-vehicular activity. However, the altered gravity environments regularly experienced during space exploration missions require sensorimotor adaptation. While nearly all astronauts eventually adapt there appears to be substantial individual differences in how quickly and effectively this adaptation occurs which are not well understood. It is thought that sensory conflict -- unexpected sensory signals -- when sustained, result in adaptation. However, even in an unchanged environment sensory signals are noisy and will lead to sensory conflict. The central nervous system must avoid inappropriately adapting to noisy sensory signals when the environment has not changed. Presumably if the conflict is consistent and sustained, the central nervous system can become more certain the conflict is due to a change in the environment and not created by sensory noise. Thus, we hypothesize that vestibular perceptual "noise" is a limiting factor in sensorimotor adaptation. If noise levels are high, a more sustained sensory conflict is required before adaptation is appropriate. We have extensive experience measuring perceptual "noise" through vestibular perceptual thresholds (i.e., how small of a stimulus can be correctly identified). Substantial inter-individual differences exist in vestibular perceptual thresholds which may help explain the poorly understood differences in altered gravity adaptation. Thus we aim to experimentally test whether individuals with higher "noise" levels, as measured by perceptual thresholds, are indeed more challenged to adapt to an altered gravity environment. A hyper-gravity environment on our Eccentric Rotator centrifuge is utilized as an altered gravity test-bed to study adaptation. Future experiments may extend to study hypo- and microgravity adaptation environments. Operationally relevant tilt perception and manual control tasks will serve as metrics to track adaptation. The hypothesis will be tested in a series of three experiments in which the adaptation stimulus will advance from being highly controlled by the experimenter to more operationally relevant. We aim to deliver a prediction tool to identify individuals which may have difficulty adapting to altered gravity environments and may benefit from sensorimotor training. The tool will require less than one hour of test time and will involve benign vestibular perceptual threshold tests. The proposed set of experiments will greatly enhance our understanding of the drivers and limiting factors for sensorimotor adaptation to altered gravity environments, and may inform the development of sensorimotor training countermeasures.

In the past year we have undergone an extensive investigation of the intra-subject variability in perceptual noise level, as assayed with vestibular perceptual thresholds. While substantial inter-individual differences (10x in normal, healthy subjects) are known to exist in vestibular perceptual thresholds, little is known how an individual's threshold level varies over repeated testing or over time. If individual differences in sensory noise levels are predictive of individual differences in sensorimotor adaptability, it is important to understand how consistent an individual's noise level remains and over what time frame. We have measured vestibular perceptual thresholds (yaw rotation and roll tilt) in 4 subjects repeatedly over the time course of 7 months. Preliminary results show that all 4 of our subjects exhibit remarkably consistent thresholds, suggesting that vestibular perceptual noise level is an innate characteristic, at least over the time course of several months. Future research will continue to investigate the relationship between an individual's sensory noise level and their ability to adapt to altered gravity environments.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

There are several Earth-based impacts of the current research project. First, while predicting an individual's capacity for sensorimotor adaptation is important for astronauts on orbit, it is also critical for sensorimotor rehabilitation patients here on Earth. Being able to predict a patient's capacity for sensorimotor adaptation may allow for individualized rehabilitation plans. Second, our advances in measuring vestibular perceptual thresholds may lead to a powerful diagnostic tool in the clinic. Current practices rely almost entirely on vestibular occular reflex (VOR) responses to diagnose patients complaining of dizziness or vertigo. However, since the symptoms (dizziness and vertigo) are illusory perceptions these reflex response tests may leave patients undiagnosed. Using vestibular perceptual thresholds as part of the test battery may provide improved diagnostics in the clinic here on Earth.

Task Progress:	We aim to investigate the relationship between vestibular perceptual "noise" and sensorimotor adaptation. Specifically we hypothesize that noise is a limiting factor for adaptation such that an individual's noise level, as measured using vestibular perceptual thresholds, will be predictive of their ability to adapt to altered gravity environments. To address this hypothesis, we have made extensive progress in two distinct areas. First we have made substantial technical progress on modifying our Eccentric Rotator device to provide the required capability for the proposed hyper-gravity experiments. This has included adding a dedicated wi-fi network to directly transmit subject responses to our device control system, developing software to allow for our proposed manual control experiments, and safety testing the device with these modifications for the conditions we aim to test. These technical enhancements are essential for providing the capability for our proposed hyper-gravity sensorimotor adaptation experiments. Second, we have made scientific and methodological progress on quantifying vestibular perceptual "noise" by threshold measures. Methodologically, we have implemented a novel two-stage reporting task for subjects when measuring thresholds. In the task, after the motion for each trial the subject first responds which direction they think they moved (left or right) and then reports if they were certain in this response or not. This novel task allows us to conduct threshold estimate analysis using the traditional binary, signal-detection framework and/or an alternative trinary model that includes an "indecision boundary." By utilizing the two-stage task we are able to gain a better understanding of the decision making process behind vestibular perceptual thresholds as well as improve our threshold estimates. Scientifically, we have conducted an extensive study investigating the intra-individual variability in vestibular perceptual thresholds. Over the time course of 7 months, our preliminary results suggest that an individual's threshold, and thus "noise" level, is remarkably consistent. This lack of variability in an individual's noise level makes an individual's threshold a potentially operationally useful predictor of capability for sensorimotor adaptation. In the future, we aim to further investigate the relationship between vestibular perceptual noise and sensorimotor adaptation by correlating our threshold measures with an individual's performance in our hyper-gravity adaptation experiments.
Bibliography Type:	Description: (Last Updated: 11/28/2025)
Abstracts for Journals and Proceedings	Beckers NWM, Young LR, Karmali F, Clark TK. "Studying the Innate Capacity for Sensorimotor Adaptation to Altered Gravity Levels." 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. http://www.hou.usra.edu/meetings/hrp2014/pdf/3225.pdf , Feb-2014
Abstracts for Journals and Proceedings	Clark TK, Newman MC, Oman CM, Merfeld DM, Young LR. "Human Perception of Roll Tilt in Hyper-Gravity: Experiments and Modeling." XXVIIIth Barany Society Meeting, Buenos Aires, Argentina, May 25-28, 2014. XXVIIIth Barany Society Meeting, Buenos Aires, Argentina, May 25-28, 2014. , May-2014
Abstracts for Journals and Proceedings	Clark TK, Yongwoo Y, Galvan-Garza RC, Bernudez-Rey MC, Merfeld DM. "How Many Decision Boundaries Contribute to Human Vestibular Decisions?" Neuroscience 2014 (Society for Neuroscience Meeting), Washington, DC, November 15-19, 2014. Neuroscience 2014, Washington, DC, November 15-19, 2014. Program#/Poster#: 825.01/JJ13. Available at: http://www.abstractsonline.com/Plan/ViewAbstract.aspx?sKey=26700404-e328-49bc-9a8d-1ff790fc39a0&cKey=98943f99-98c4-4dde-b4aa-8c192767332e&mKey=54c85d94-6d69-4b09-afaa-502c0e680ca7 ; accessed 9/1/2017. , Nov-2014
Abstracts for Journals and Proceedings	Galvan RC, Bloomberg JJ, Mulavara AP, Clark TK, Merfeld DM, Oman CM. "Improving Sensorimotor Function and Adaptation using Stochastic Vestibular Stimulation." 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. http://www.hou.usra.edu/meetings/hrp2014/pdf/3102.pdf , Feb-2014
Abstracts for Journals and Proceedings	Newman MC, Clark TK. "Methods for Studying Human Orientation Perception and Control in Hyper-Gravity." 85th Annual Scientific Meeting, Aerospace Medical Association, San Diego, CA, May 10-15, 2014. Aviation, Space, and Environmental Medicine. 2014 Mar;85(3):265. See http://www.ingentaconnect.com/content/asma/asem/2014/00000085/00000003 , Mar-2014
Articles in Peer-reviewed Journals	Clark TK, Newman MC, Oman CM, Merfeld DM, Young LR. "Human perceptual overestimation of whole body roll tilt in hypergravity." J Neurophysiol. Epub 2014 Dec 24. PubMed PMID: 25540216 , Dec-2014
Awards	Clark TK. "2014 Stanley Roscoe Award for Best Doctoral Thesis from the Aerospace Human Factors Association (AsHFA). Thesis title: 'Human Perception and Control of Vehicle Roll Tilt in Hyper-Gravity.' May 2014." May-2014
Papers from Meeting Proceedings	Clark TK, Newman MC, Merfeld DM, Young LR. "Pilot control and stabilization of a rate-controlled vehicle in hyper-gravity." 2014 IEEE Aerospace Conference, Big Sky, MT, March 1-8, 2014. In: 2014 IEEE Aerospace Conference, Digest of Papers, Paper MVL14.01, 8 p. http://dx.doi.org/10.1109/AERO.2014.6836299 , Mar-2014